

Packed Levitated Marker for Entity and Relation Extraction (PL-Marker)

자연어처리팀 김유진, 이기성, 정천감, 조해창

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PL-Marker

- Span representation method for Named Entity Recognition (NER), Relation Extraction (RE)
- Consider **interrelation between spans**
by strategically packing markers (solid & levitated) in encoder
- SOTA in RE on ACE 2005; SOTA in Joint Entity and RE on SciERC; #2 in NER on Ontonotes v5 (*[paper with code](#)*)

Span Representation?

- "Contiguous random spans (pairs)" instead of "random tokens (words)"
 - e.g.) Super Bowl 50 was an American football game to determine the champion
↳ Span
- BERT: predict a token after masking tokens
- SpanBERT: predict a span after masking spans

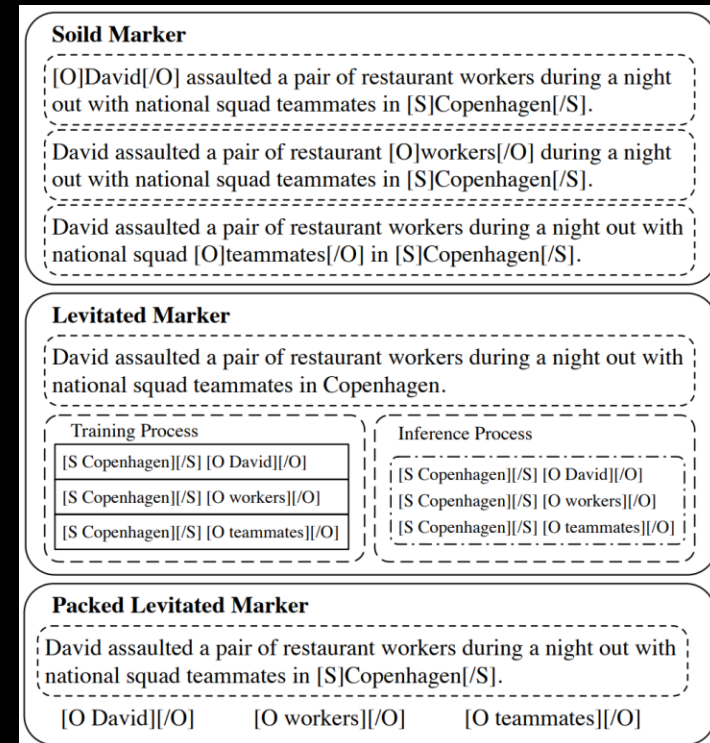
3 Span Representation Methods

1. **Span pre-training**: Enhance span representation for PLMs via **span-level pre-training tasks**
2. **Knowledge infusion**: Infuse **external knowledge** (knowledge graph, synonym net, Wikipedia anchor texts) into their models
3. **Structural extension**: **Add reasoning modules** (Biaffine attention, graph propagation, memory flow) to existing models
 - PURE (Zhong and Chen), PL-Marker (Fusion subject-oriented packing scheme)

Packed-Levitated Marker?

Span representation extraction

- **Solid marker:** Explicitly insert two solid markers (start, end)
BUT Cannot handle overlapping span pairs at a time
- **Levitated marker:** Share same start/end position with tokens while keeping position id of original tokens unchanged
BUT Ignore interrelation between spans
- **Packed levitated marker:** Pack all objects for same subject into an instance



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Pack it wisely!

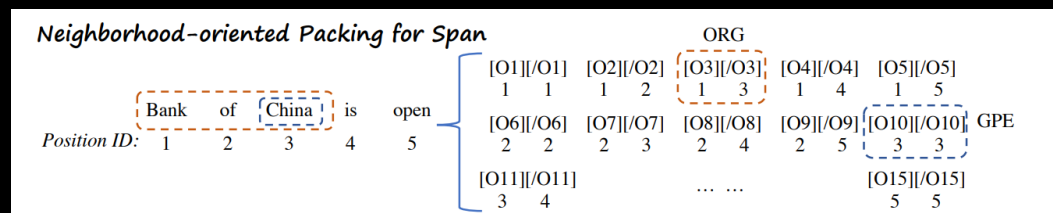
- Packing all span pairs together is costly...
 - No. of levitated marker $\uparrow$ -> **PLM complexity** $\uparrow \uparrow \uparrow \uparrow$
- For span classification tasks, divide spans into several groups (neighbor spans); Neighborhood-oriented packing strategy & levitated marker
- For complicated span pair classification tasks, subject-oriented packing strategy & fusion of solid/levitated markers
 - Solid marker for subject spans / levitated markers for object spans

* Span classification tasks: Assigning a label to 1 span of text

* Span pair classification tasks: Identifying & classifying relationship between 2 spans of text

Neighborhood-oriented Packing

- A large number of candidate spans -> **group markers into K batches** for higher speed & practical feasibility
- **Neighbor spans** (e.g. span with the same start token)
 - Example of neighbors: "Bank", "Bank of", ..., "Bank of China is open"



- Candidate spans set $S(X) = \{(1, 1), (1, 2), \dots, (\lceil \frac{K}{L} \rceil, K - \lfloor \frac{K-1}{L} \rfloor * L)\}$
- Span representation: $\psi(s_i) = [h_a^{(s)}; h_b^{(e)}]$ (Start & end token marker)

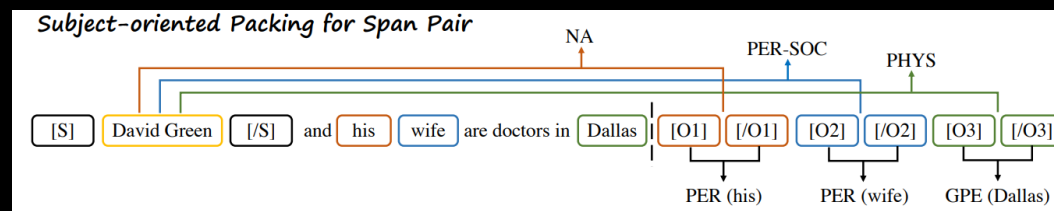
Subject-oriented Packing

- Bind a **subject span** with **solid markers** [S] & [/S], candidate **object spans** with **levitated markers** [O] & [/O], then pack them into an instance

- Modified sequence with markers: $\hat{X} = \dots[S], x_a, \dots, x_b, [/S], \dots, x_{c_1} \cup [O1], \dots, x_{d_1} \cup [/O1], \dots, x_{c_2} \cup [O2], \dots, x_{d_2} \cup [/O2], \dots$

- Span pair representation: $\phi(s_i, s_j) = [h_{a-1}; h_{b+1}; h_c^{(s)}; h_d^{(e)}]$

- Impaired emphasis on object span due to levitated markers
-> Inverse relation from object to subject for bidirectional prediction



Solid & Levitated Marker?

- Each pair of levitated markers is tied by **directional attention**
 - Binding 2 pairs of levitated markers -> markers **failing to identify their partners**
- For complex task: Use a fusion of solid & levitated markers

Complexity Analysis

- In general, leviated marker -> Sequence length $\uparrow$ -> Computation $\uparrow \uparrow \uparrow \uparrow$
- **Span classification tasks:** Markers are **grouped into several batches** to control sequence length -> Complexity increases **nearly linearly**
- **Span pair classification tasks:** **Small number** of candidate spans -> **Limited** computation increase

QUESTION?



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Experimental Setup

• Dataset

- NER: Flat (CoLNN03, OntoNotes 5.0, Few-NERD) & nested
- End-to-end RE: Nested datasets (ACE04, ACE05, SciERC)

• Split

- ACE04 into 5 folds, ACE05 into train, development, test sets
- Official split for other datasets

• Evaluation metrics

- NER: Span-level eval (boundary, type)
- RE: Boundaries eval ("Rel", boundary, relation) & strict eval ("Rel+", boundary, relation, type)

• Implementation details (256 group size, 5 seeds, average score reported)

- RoBERTa encoder for flat NER / BERT-base-uncased for ACE04 / ALBERT-xxlarge-v1 for ACE05 / SciBERT-scivocab-uncased for SciERC

Dataset	#Sents	#Ents (#Types)	#Rels (#Types)
CoNLL03	22.1k	35.1k (4)	-
OntoNotes 5.0	103.8k	161.8k (18)	-
Few-NERD	188.2k	491.7k (66)	-
ACE05	14.5k	38.3k (7)	7.1k (6)
ACE04	8.7k	22.7k (7)	4.1k (6)
SciERC	2.7k	8.1k (6)	4.6k (7)

Table 1: The statistics of the adopted datasets.

NER

- **Neighborhood-oriented packing** outperforms, especially on **longer sentences** -> Better model **interrelation among neighbor spans**
 - Few-NERD: avg. 325 candidate spans (90, 174 for CoNLL03, OntoNotes 5.0)
- Improved F1 over T-Concat on all benchmarks -> **Levitated markers' advantage** in aggregating span representation for entity type prediction
 - T-Concat: Concatenate span's boundary (start/end) token representation
- Outperforms SeqTagger -> Effective in interrelation btw diverse types of entities
 - SeqTagger: NER as seq tagging task. Apply token-level classifier

Model	CoNLL03	OntoN5	F-NERD
Ma and Hovy (2016)	91.0	86.3	-
Devlin et al. (2019)	92.8	89.2	68.9
Li et al. (2020)	93.0	91.1	-
Yu et al. (2020)	93.5	91.3	-
Yan et al. (2021)	93.2	90.4	-
SeqTagger (Our impl.)	93.6	91.2	69.0
T-Concat (Our impl.)	93.0	91.7	70.6
Random Packing	93.9	91.8	61.5
PL-Marker (Our model)	94.0	91.9	70.9

Table 2: Micro F1 on the test set for the flat NER. OntoN5: OntoNotes 5.0; F-NERD: Few-NERD.

RE

- Outperforms by 1.7-2.5%p on BERT
- Best on SciBERT
- Both NER, RE improved using ALBERT
- Compared to PURE (Full), 4.1-4.3%p ↑
 - Thanks to **modeling interrelation btw same subject/object entity pairs in training**
 - PURE (Full): 2 pairs of solid markers
 - PURE (Approx): 2 pairs of levitated markers

Model	Encoder	Rep Type	ACE05			ACE04			SciERC		
			Ent	Rel	Rel+	Ent	Rel	Rel+	Ent	Rel	Rel+
Li and Ji (2014)	-	-	80.8	52.1	49.5	79.7	48.3	45.3	-	-	-
SPTree (Miwa and Bansal, 2016)	LSTM	T	83.4	-	55.6	81.8	-	48.4	-	-	-
DYGIE (Luan et al., 2019) [◇]	ELMo	T	88.4	63.2	-	87.4	59.7	-	65.2	41.6	-
Multi-turn QA (Li et al., 2019)	BERT _L	-	84.8	-	60.2	83.6	-	49.4	-	-	-
OneIE (Lin et al., 2020)		T	88.8	67.5	-	-	-	-	-	-	-
DYGIE++ (Wadden et al., 2019) [◇]	BERT _B / SciBERT	T	88.6	63.4	-	-	-	-	-	-	-
TriMF (Shen et al., 2021) [◇]		T	87.6	66.5	62.8	-	-	-	70.2	52.4	-
UniRE (Wang et al., 2021d) [◇]		T	88.8	-	64.3	87.7	-	60.0	68.4	-	36.9
PURE-F (Zhong and Chen, 2021) [◇]		S	90.1	67.7	64.8	89.2	63.9	60.1	68.9	50.1	36.8
PURE-A (Zhong and Chen, 2021) [◇]		L	-	66.5	-	-	-	-	-	48.1	-
PL-Marker (Our Model) [◇]		S&L	89.8	69.0	66.5	88.8	66.7	62.6	69.9	53.2	41.6
TableSeq (Wang and Lu, 2020)	ALB _{XXL}	T	89.5	67.6	64.3	88.6	63.3	59.6	-	-	-
UniRE (Wang et al., 2021d) [◇]		T	90.2	-	66.0	89.5	-	63.0	-	-	-
PURE-F (Zhong and Chen, 2021) [◇]		S	90.9	69.4	67.0	90.3	66.1	62.2	-	-	-
PL-Marker (Our Model) [◇]		S&L	91.1	73.0	71.1	90.4	69.7	66.5	-	-	-

Table 3: Overall entity and relation F1 scores on the test sets of ACE04, ACE05 and SciERC. The encoders used in different models: BERT_B = BERT_{BASE}, BERT_L = BERT_{LARGE}, ALB_{XXL} = ALBERT_{XXLARGE}. Specially, TriMF, UniRE, PURE and PL-Marker apply BERT_{BASE} on ACE04/05 and apply the SciBERT on SciERC. [◇] denotes that the model leverages the cross-sentence information. Representation Type: T-T-Concat; S-Solid Marker; L-Levitated Marker. Model name abbreviation: PURE-F: PURE (Full); PURE-A: PURE (Approx.).

Inference Speed

- Span model
 - **PL-Marker**: F1 $\uparrow$ but speed $\downarrow$ by 60%
 - **Two stage** (fast BASE-size T-Concat to filter candidate spans + PL-Marker): Similiar performance with 3.1x speed
 - Group size 512 -> PL-Marker slows down
- Span pair model
 - Compared to PURE (Approx.), higher F1 but less efficient
 - Compared to PURE (Full), higher F1 & 2.2x-2.8x faster
 - Effectiveness of fusion markers and packing strategy

Model	K	CoNLL03		Few-NERD	
		Ent (F1)	Speed (sent/s)	Ent (F1)	Speed (sent/s)
SeqTagger	-	93.6	138.7	69.0	142.0
T-Concat	-	93.0	137.2	70.6	126.8
PL-Marker	128	94.0	54.8	70.9	23.8
	256	-	39.6	-	25.8
	512	-	22.9	-	18.3
Two-stage	16	93.7	87.1	70.8	80.6
	32	94.0	83.3	70.9	79.8

Table 4: Micro F1 and efficiency on NER benchmarks with respect to the model and different packing group size K . We adopt a maximum span length of 8 for CoNLL03 and 16 for Few-NERD.

Model	ACE05		SciERC	
	Rel (F1)	Speed (sent/s)	Rel (F1)	Speed (sent/s)
PURE (Full)	67.7	76.5	50.1	88.3
PURE (Approx.)	66.5	593.7	48.8	424.2
PL-Marker	69.3	211.7	52.8	190.9

Table 5: Comparison of our RE model and PURE in relation F1 (boundaries) and speed. We report the result with BASE encoders. All models adopt the same entity input from the entity model of PURE.

Case study

- **Collect contextual info** (Taiwan, mainland) to predict "Cross Strait" as "ORG"
- Integrally **consider interrelation btw same-object relational facts** in training
 - Obtain both "Liana" and "her parents" are located in "Manhattan"

Named Entity Recognition

Text: This is the Cross Strait program on CCTV International Channel. ... Candidates for the giant pandas to be presented to Taiwan as gifts from the mainland may increase. ...

T-Concat: (Cross Strait, WORK OF ART), (CCTV International Channel, ORG), (Taiwan, GPE)

Our: (Cross Strait, ORG), (CCTV International Channel, ORG), (Taiwan, GPE)

Relation Extraction

Text: Liana drove 10 hours from Pennsylvania to attend the rally in Manhattan with her parents

PURE: (Liana, located in, Manhattan)

Our: (Liana, located in, Manhattan), (her parents, located in, Manhattan)

Table 6: Case study of our NER and RE model.

Ablation study

- Solid marker: 2.0%-3.8% ↓
- Inversion relation: 0.9%-1.1% ↓
- Entity type loss: 0.9%-1.1% ↓
- Type marker: 0.4%-0.7% ↑
 - Added auxiliary entity type loss to RE model to inject entity type info from NER
 - => **Entity type prediction error from NER model propagated to RE model**

Model	ACE05		SciERC	
	gold	e2e	gold	e2e
PL-Marker	74.0	69.0	72.5	53.2
w/o. solid marker	72.0	67.3	68.7	50.6
w/o. inverse relation	72.9	68.1	71.6	52.7
w/o. entity type loss	73.4	68.4	72.3	53.2
w. type marker	74.0	68.3	72.1	53.0

Table 7: The relation F1 (boundaries) on the test set of ACE05 and SciERC with different input features for the ablation study. gold: use the gold entities; e2e: use the entities predicted by our entity model. w/o.: without. w.: with.

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Conclusion

- Considered interrelation between spans & span pairs
- Obtained SOTA F1 scores on NER / RE tasks
- First to apply levitated markers on NER

QUESTION?



THANK YOU!

